MiCM McGill initiative in Computational Medicine



Intro to MATLAB



Workshop Lead: Meghana Munipalle

Date: October 21, 2026

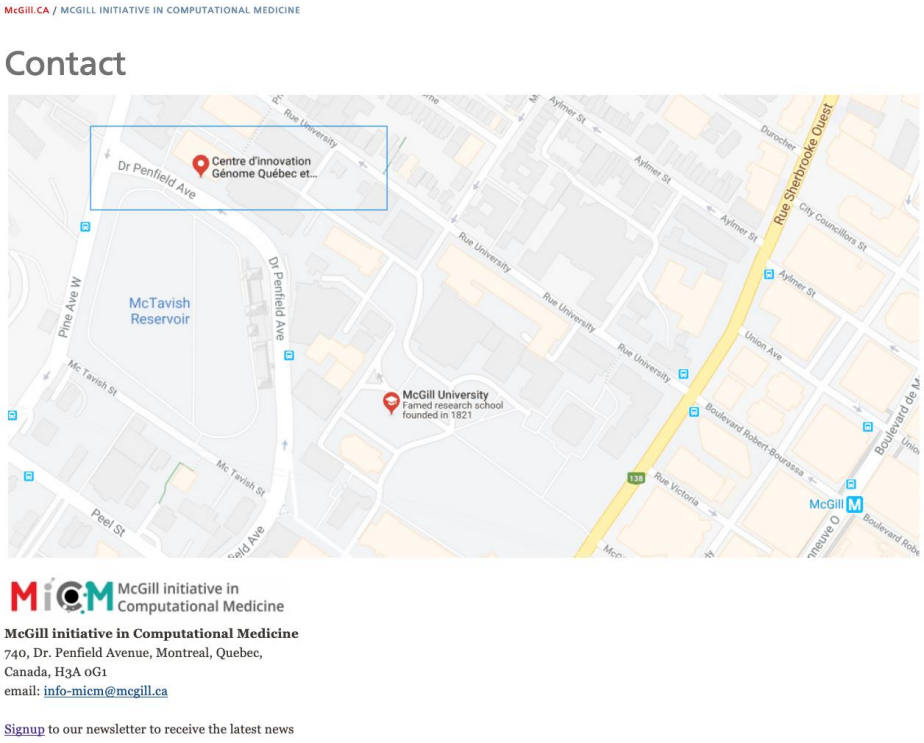
MiCM McGill initiative in Computational Medicine

McGill initiative in Computational Medicine

Mission : aims to deliver inter-disciplinary research programs and empower the use of data in health research and health care delivery

Scan the QR code to sign up for our **mailing list!**

https://www.mcgill.ca/micm



Outline

Module 1: MATLAB Basics

- Calculations on the command line
- Programming in MATLAB
- Exercises!

Module 2: MATLAB for Science

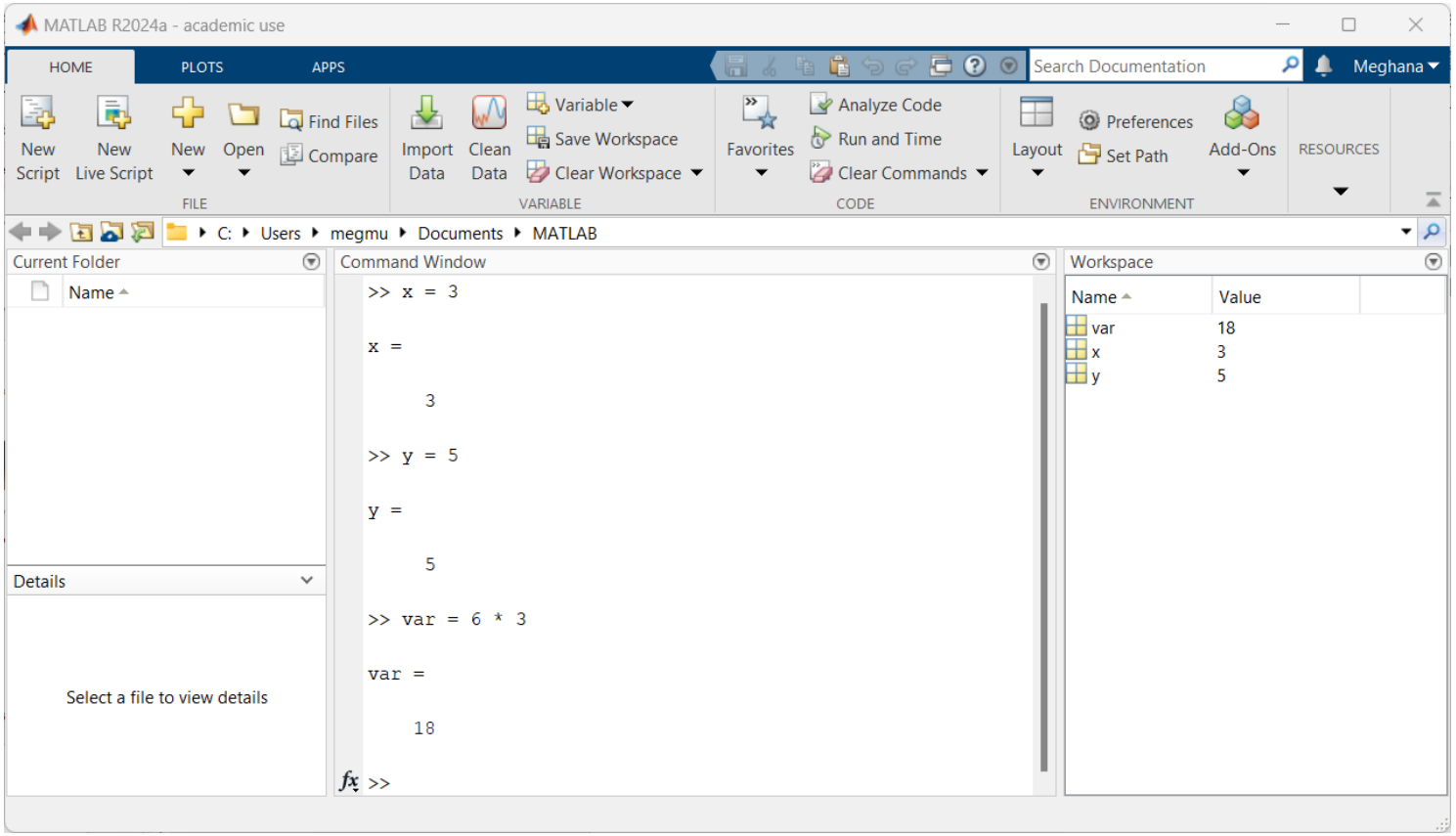
- 2D & 3D Plotting
- Numerical Methods
- Exercises!

Module 3: Additional Resources!



Getting Started

- Tour of the environment



Code Along

The command line

- Using MATLAB as a fancy calculator
- Vectors, matrices, and arrays
- Basic built-in functions



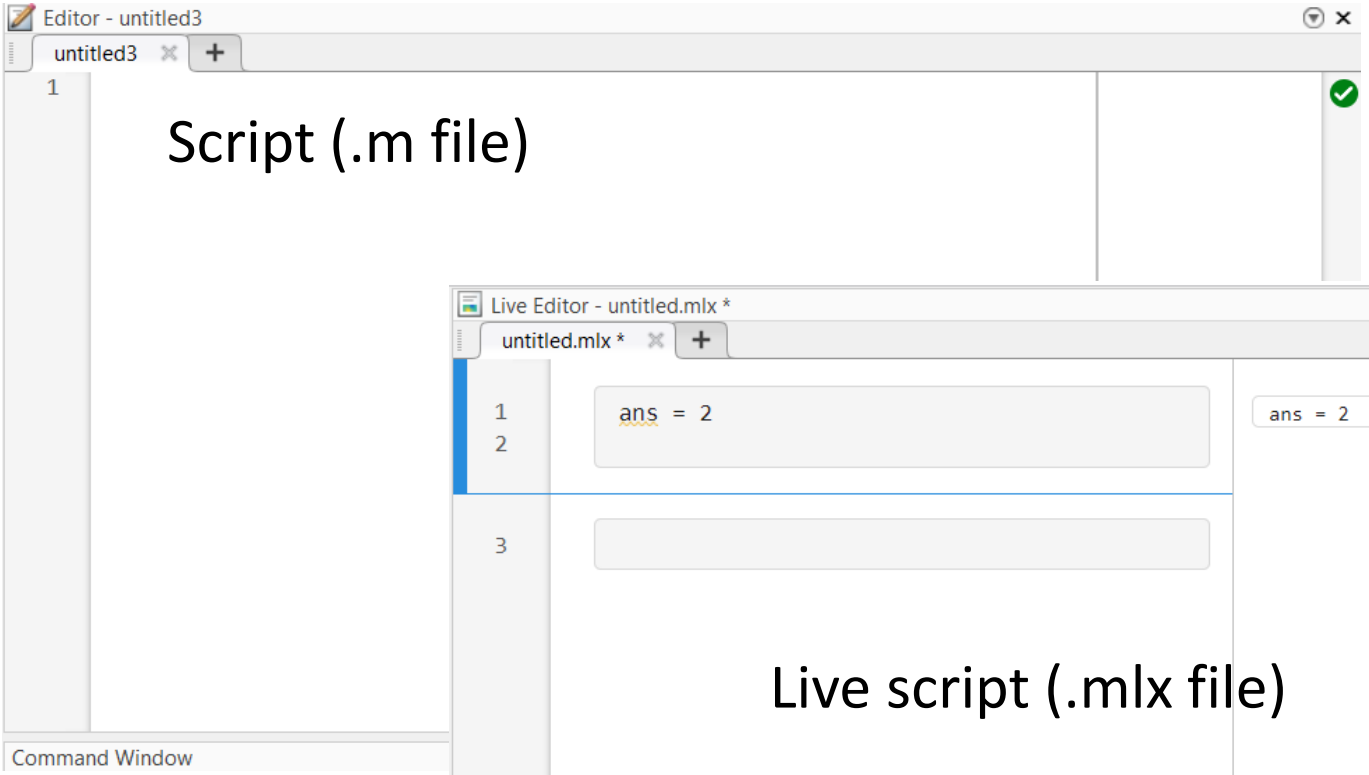
Try It Out

Exercise 1: Command Line

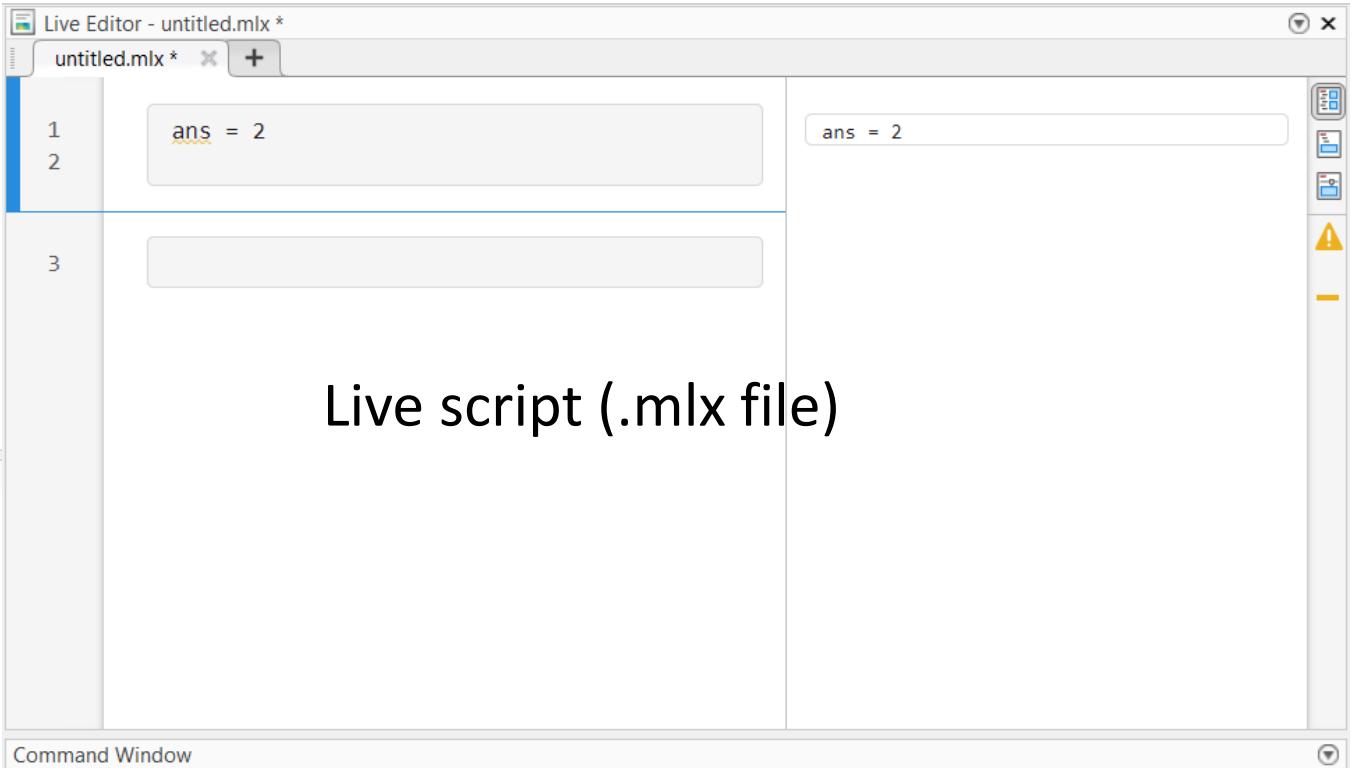
1. Create a row vector of odd numbers through 21 (without simply listing all the numbers):
   - L = 1 3 5 7 9 11 13 15 17 19 21
2. Find the sum of all elements in this vector using 1 command:
   - [1, 3, -5, 24, 8, -76, 44, -2, 7, 13]
3. Calculate the hypotenuse of a triangle with side lengths 12 and 15 (try in 1 line!)
4. Type the following lines, pressing enter in between each
   - p = (3*2)^2
   - q = (3*2)^2;
   - What is the difference between the outputs?
5. Form a 6 x 1 vector **a** of zeros using the zeros command

Code Along

MATLAB Scripts



Script (.m file)



Live script (.mlx file)

Code Along

MATLAB Scripts

- Variables and functions
- If-statements and loops
- Text and characters
- Basic errors and troubleshooting

Try It Out

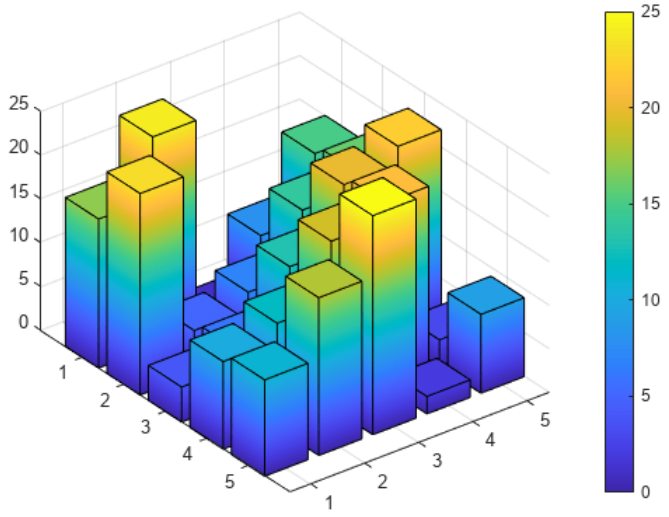
Exercise 2: Programming

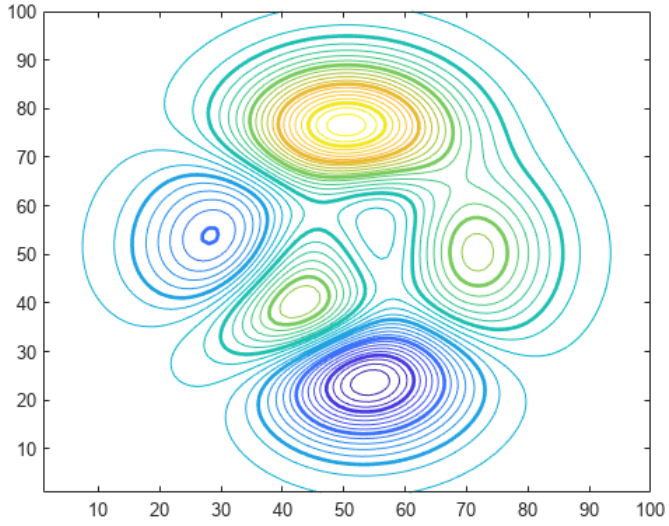
1. Create functions for the different tasks in the live script!
2. Create a function that calculates the sum S of elements:
$$a_i = \sqrt{2i - 1} \qquad i = 1, 2, \ldots$$
3. Write a function called div5 that takes a number as input and checks to see if it is divisible by 5 (and then print a message that indicates the result).
4. The $n \times n$ Hilbert matrix H has the entries $H_{i,j} = 1/(i + j - 1)$, $i, j = 1, 2, \ldots, n$. Create a nested for-loop to generate the 5x5 Hilbert matrix and use the built-in function **hilb** to check your answer.
5. Generate a string consisting of *n* spaces (multiple ways to do this!)
6. Sam gets paid compound interest at a rate defined at 5% per annum. Calculate his resulting investment each year and after 10 years. Change the rate on the 8th year to 5.75%.

Code Along

2D & 3D Plotting

- Basic plotting
- Interacting with files

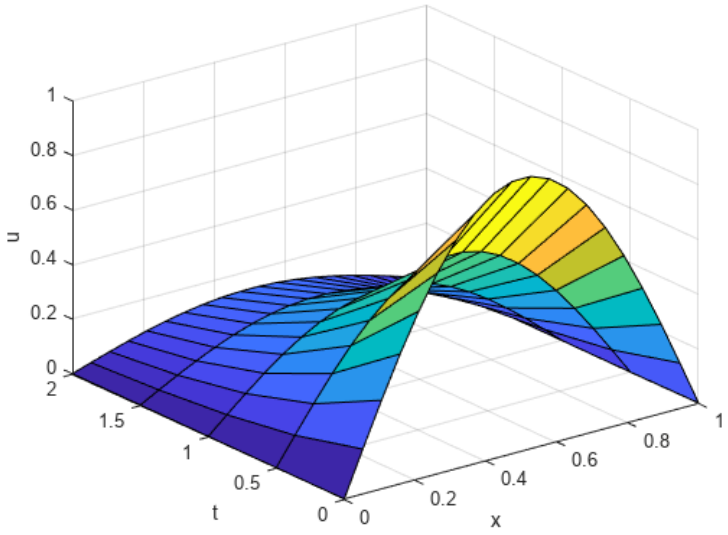




Code Along

Some useful numerical methods

- Root finding
- Solving differential equations
- Curves of best fit

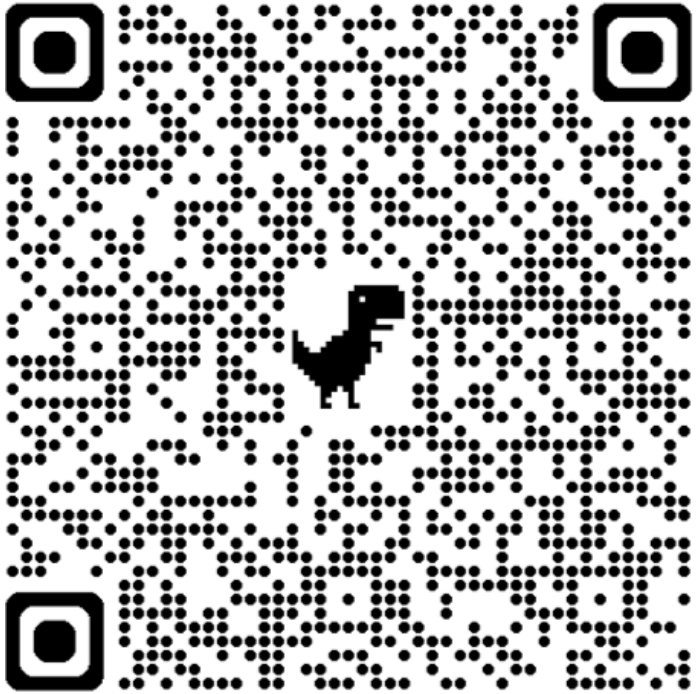


Try It Out

Exercise 3: Data Vis & Analysis

1. Work through activity3_qpcr.mlx
2. Work through activity3_penguins.mlx

Let us know what you think!



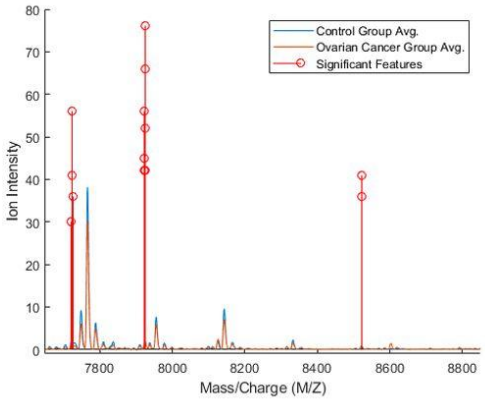
Quick Links!

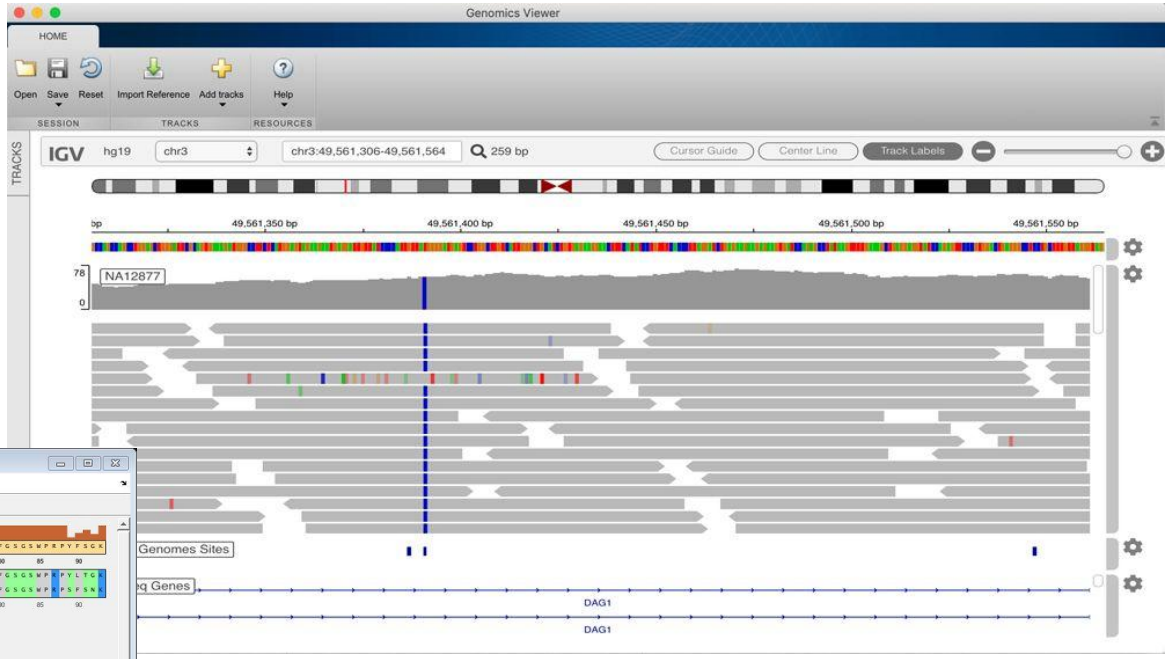
- Want to learn more?
  - MathWorks self-paced courses
  - LinkedIn Learning – free through your student account

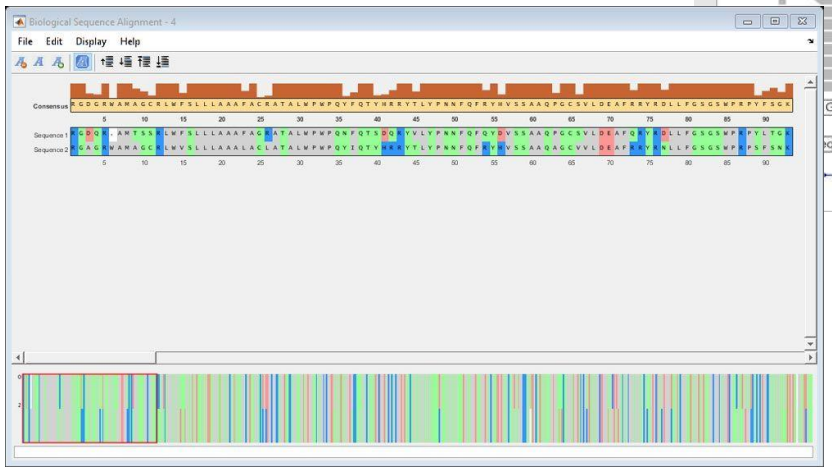
- Want to get help?
  - https://www.mathworks.com/help/matlab/ Main documentation- very extensive
  - StackOverflow- a must ☺

Tools for Computational Medicine

- Bioinformatics Toolbox (option when installing)

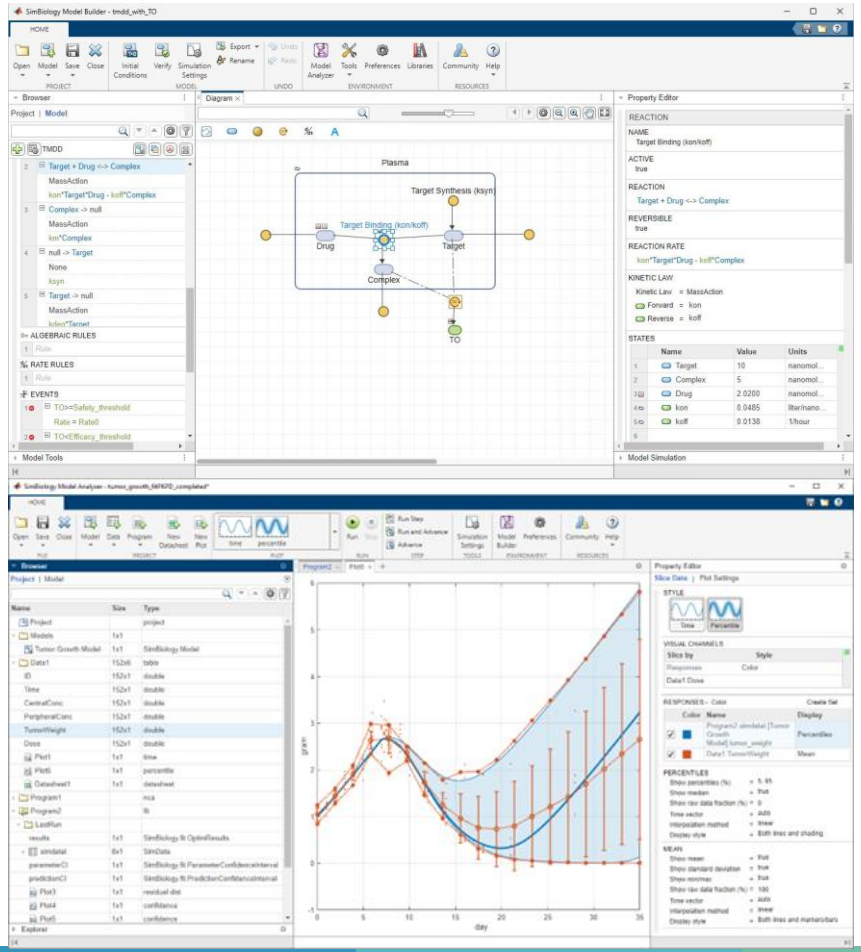


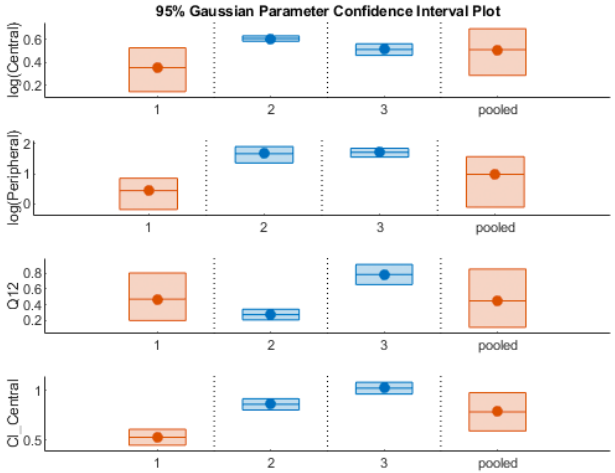




Tools for Computational Medicine

- SimBiology (Simulink product)





ODE-based models (for pharmacokinetics, etc), Monte Carlo simulations, parameter estimation, tons of analysis!

To summarize

Now you are ready to:

- Use MATLAB for many different types of calculations
- Navigate the MATLAB command line and live scripts
- Input, plot, and analyze data
- Find tutorials and toolboxes specific to your research and/or interests!

CODE -TO- MANUSCRIPT BOOTCAMP

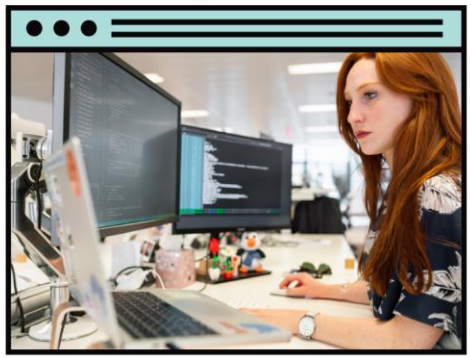
October 26-30
McIntyre Medical

Ready to turn your research into a paper, but not sure where to start?

Join the QLS-MiCM, CDSI, D2R, and C3G for a hands-on bootcamp where you'll learn how to **visualize biological/ omics data** and prepare a manuscript.

Get started with simple, customizable techniques for data visualization and writing in **R**, **Python**, **Inkscape**, and **LaTeX**!

Register now!



Sign up here:

Canadian Centre for Computational Genomics

McGill initiative in Computational Medicine

Thank you for attending!

Let us know what you think!

